Arogya Relay Live Demo Guide

Zero-Loss Rural Telehealth · 2G & Offline-Resilient Distributed Architecture

System Documentation
Date: October 2026
Version: v2.4 (Branch-Isolated)

1. What Was Upgraded (Recent Architectural Fixes)

Clean Role Separation (Zero Cross-Role Bleed):

- 👤 **Citizen / Patient Portal:** Completely isolated for citizens. Removed all doctor case queues, prescribing forms, chemist inventory, and the confusing "Switch to Staff Mode" button. Citizens now only see health guidance, consultation intake, and their prescription wallet.
- 👨‍⚕️ **Doctor Workstation:** Directly opens the live Clinical Cases Queue, AI-powered Zero-Loss Care Guidance, and prescription authoring with cryptographic digital signing authority.
- 🏪 **Chemist Workstation:** Stays locked to the Jan Aushadhi Dispensary counter. Removed the "Back to Dashboard" loop that dumped pharmacists into doctor triage screens.
- 👩‍🌾 **ASHA Field Worker:** Streamlined for rural health camps, voice vitals dictation, lesion camera capture, and offline local-first synchronization.
- **Consolidated Single Logout:** Removed all conflicting duplicate logout buttons from hero cards and sidebars. Standardized a single universal `🚪 Log Out` button in the topbar.
- **Synchronized Session State:** Eliminated the phantom "Sign In / Sign Up" button that previously appeared while authenticated in offline mode.

2. Platform User Accounts & Demo Credentials

All accounts configured in the active platform directory (visible in the System Administration Console):

#	USER / DISPLAY NAME	ROLE BADGE	LOGIN EMAIL ADDRESS	PASSWORD	PRIMARY SCOPE / INTERFACE
1	**Pratyush Kiran Rath**	ADMIN	`pratyushkiranrath4@gmail.com`	`admin123`	System Governance, Doctor NMC verification, Escrow audit
2	**Dr. Rajesh Patel, MBBS**	DOCTOR	`dr.patel@ruralhealth.org`	`demo123`	Verified Physician, Clinical Triage, Digital Rx Signing
3	**Dr. Ananya Sharma, MD**	DOCTOR	`dr.ananya@health.gov.in`	`demo123`	Verified NMC Doctor, Paediatrics & General Medicine
4	**Priya Devi (ASHA)**	HEALTH WORKER	`asha.priya@field.arogya.org`	`demo123`	Offline Field Screening, Voice Vitals Dictation, Camp Sync
5	**Jan Aushadhi Kendra #1084**	CHEMIST	`jan.aushadhi.kendra@pharma.org`	`demo123`	QR Prescription Scanner, Generic Substitution, Token Burn
6	**Ramesh Soren**	PATIENT	`citizen.ramesh@gmail.com`	`demo123`	Citizen Health Pass, Disease Library, Medication Alarms
7	**Milanium Biswal**	PATIENT	`mileniumbisswal10042008@gmail.com`	`demo123`	Citizen Health Pass, Teleconsultation intake

⚡ **1-Click Fast Switching:** On the home landing page, click any of the "⚡ **Quick Demo Accounts**" buttons located right below the Sign In form to auto-fill these credentials instantly without typing.

3. Step-by-Step Live Demo Presentation Script

To demonstrate the real-world value of Arogya Relay, execute the live presentation across **5 sequential acts** representing the full rural healthcare lifecycle:

ACT 1: The Citizen Experience — Low Literacy & Multilingual Access

Step 1 of 5

Login: Sign in as `citizen.ramesh@gmail.com` (Password: `demo123`).

- **Role Recognition:** Show that the system immediately displays `Active Citizen Session`. Point out the total absence of clinician jargon or back-office buttons.
- **Arogya Gyan (Disease Library):** Open `📖 Health Library`. Switch the topbar language to **Hindi** or **Odia** to show localized home remedies, nutrition advice, and emergency danger signs.
- **Prescription Wallet & Alarms:** Click `💊 My Prescriptions & Reminders`. Show the patient's personalized dose reminders, compliance alarms, and their tamper-proof **Digital Prescription QR Pass**.
- **Logout:** Click `🚪 Log Out` in the topbar to cleanly exit.

ACT 2: The ASHA Field Worker — Offline Screening & AI Vitals Dictation

Step 2 of 5

Login: Sign in as `asha.priya@field.arogya.org` (Password: `demo123`).

- **Field Context:** Explain that Priya Devi conducts outreach in hamlets with poor cellular coverage.
- **Voice Vitals Dictation:** Launch `+ New Clinical Screening Workstation`. Click the microphone icon and speak symptoms or vitals; demonstrate real-time structured parsing of temperature, pulse, and SpO2.
- **Drishti AI Visual Lesion Camera:** Show how uploading a dermatological lesion photo instantly matches clinical reference cases.
- **Save Screening:** Save the assessment. Point out the offline queue badge showing that records cache safely locally until connectivity returns.
- **Logout:** Click `🚪 Log Out` in the topbar.

ACT 3: The Doctor — Tele-Triage & Verified Digital Prescriptions

Step 3 of 5

Login: Sign in as `dr.patel@ruralhealth.org` (Password: `demo123`).

- **Doctor Workstation:** Point out that Dr. Patel lands directly in the `◎ Cases Queue`.
- **Case Evaluation:** Open the incoming screening case submitted by the ASHA worker. Review SpO2 and pulse readings categorized by urgency (Emergency vs Urgent vs Routine).
- **Zero-Loss Care Guidance:** Show AI-assisted clinical protocols and contraindication cross-checks.
- **Digital Prescription Authoring:** Navigate to `💊 Prescriptions`. Add generic medications (e.g. Amoxicillin 500mg, Paracetamol 650mg) and demonstrate the ✓ **Verified NMC Prescriber** digital cryptographic signature.
- **Logout:** Click `🚪 Log Out` in the topbar.

ACT 4: The Jan Aushadhi Chemist — Generic Substitution & Escrow Burn

Step 4 of 5

Login: Sign in as `jan.aushadhi.kendra@pharma.org` (Password: `demo123`).

- **Dedicated Pharmacy Counter:** Notice how the interface opens straight into the `🏪 Jan Aushadhi Dispensary Workstation`.
- **Prescription Token Verification:** Enter or scan the 6-character prescription token (e.g., `PT-7821` or click Verify).
- **Generic Cost Savings:** Highlight the comparison showing how expensive branded drugs are substituted for government Jan Aushadhi generics, saving rural families 70% to 85% on medicine costs.
- **Single-Use Settlement:** Click **Dispense Medicines & Settle Escrow**. Show the token burning to prevent double-dispensing, alongside automated inventory deduction.
- **Logout:** Click `🚪 Log Out` in the dedicated chemist header.

ACT 5: The Administrator — Governance & Compliance Audit

Step 5 of 5

Login: Sign in as `pratyushkiranrath4@gmail.com` (Password: `admin123`) or visit `/admin`.

- **Platform User Directory:** View all registered users (matching the live database table).
- **Role Governance:** Change a user's role on the fly with immediate Row-Level Security policy updates.
- **Doctor NMC Verifications:** Review physician medical council registrations and license documents.
- **Audit Trail & Health:** Show the telemedicine escrow ledger, encryption status, and database health.

4. Core Technical Architecture & Safeguards

Key Innovations in Arogya Relay:

1. **Offline-First Synchronization:** IndexedDB and LocalStorage vault architecture ensure full operational capacity in zero-connectivity environments. Transactions sync automatically once 2G or broadband resumes.
2. **Zero-Loss Escrow Ledger:** Medical prescription tokens are single-use cryptographically verified primitives. Once dispensed by a Jan Aushadhi chemist, the token is invalidated across the national network to eliminate counterfeit re-fills.
3. **Non-Recursive Supabase RLS:** Stored security procedures (`public.is_admin()` , `public.is_clinician()`) execute via `SECURITY DEFINER` to bypass recursive policy loops.
4. **Multilingual Speech Dictation:** Native Web Speech API integrated with local parsing rules extracts patient vitals from conversational regional speech.

5. Presenter Checklist & Live Demo Tips

- **Dual-Browser Setup:** Open one normal window (Doctor Workstation) and one Incognito window (Chemist or Citizen) side-by-side to showcase real-time multi-actor handoffs.
- **Simulating 2G/Offline:** Open Chrome Developer Tools > Network Tab > set Throttling to **"Offline"**. Demonstrate that screening intake and disease search continue to operate seamlessly.
- **Presentation Timing:** Recommended duration is **6 to 8 minutes**, allocating roughly 1.5 minutes per role.